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CSC 34200
Hw 1 Decimal to Binary to Decimal
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Objective: we are tasked to create a program that converts base 10 numbers into binary and back to base 10. We will also determine which bit is the most and least significant bit.

C++ Code:

```cpp
/******************************************************************************
Create a program that converts base 10 numbers into binary (base 2) and then
determine which bit is the most significant. Due today by midnight
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*******************************************************************************/
#include <iostream>
#include <math.h>
using namespace std;
int main()
{
    int number,BackToNum;
    string binarystring;
    cout<<"Give a number to be converted to binary"<<endl;
    cin>>number;
    cout<<endl;
    BackToNum=0;
    //for(int i=size-1;i>=0;i--) for given  bit size
    while(number>0){
        binarystring+= to_string(number % 2 );
        number= number/2;
    }//in reverse


    string revBinarystring=string(binarystring.rbegin(),binarystring.rend());
    cout<<revBinarystring<<endl;
    cout<<"Most Significant bit: "<<revBinarystring[0]<<endl;
    cout<<"Least Significant bit: "<<revBinarystring[revBinarystring.length()-1]<<endl;

    int length=revBinarystring.length();

    for(int i =length-1; i>=0;i--){
        BackToNum+= stoi(binarystring.substr(i,1))*pow(2,i);
    }
    cout<< "Back to Base 10: "<<BackToNum<<endl;
    return 0;
}
```

```
Give a number to be converted to binary
37

100101
Most Significant bit: 1
Least Significant bit: 1
Back to Base 10: 37
```

Explanation: in order to make the number into binary we must find the remainder when dividing the number by 2 which is the base for binary and add it to a string to be able to represent it. When doing so the binary string is in reverse so we reverse the string to get the true value. To get the binary back to base 10 number we multiply 2 to the power of i times the digit of the binary string at i making back to base 10.